

Multi-Projector Dynamic Spatially Augmented Reality on Deformable Surfaces

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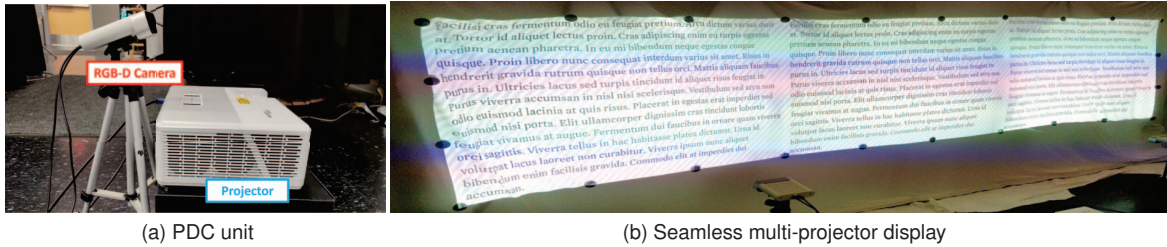


Figure 1: (a) A single Projector-Depth camera pair, or PDC. (b) A seamless multi-projector display using multiple PDCs.

ABSTRACT

Current multi-projector spatially augmented reality systems are limited to static and/or rigid surfaces. We will demonstrate the first multi-projector spatially augmented reality system on non-rigid surfaces like deformable fabrics and human body. Using two projectors and two RGB-D cameras, our system reconstructs the changing surface shape in real-time and adapts the projection from both projectors automatically to ensure a seamless, registered projection display. Such systems have applications in various domains such as art, design, entertainment and medicine. In particular, we will demonstrate two applications that showcase the flexibility and scale of the system: projection on a large deformable fabric, and a surgical guidance system for cleft palate surgery.

Index Terms: Computing Methodologies—Artificial Intelligence—Computer Vision—Image and Video Acquisition

1 INTRODUCTION

Spatially Augmented Reality (SAR) systems use projectors to illuminate physical objects for displaying digital content that can be viewed without any handheld or wearable devices such as tablets or HMDs. Such systems can be used to project on stationary surfaces (Static SAR) or non-rigid surfaces (Dynamic SAR). A Dynamic-SAR (D-SAR) system is built by one or more projectors illuminating the display surface with realtime feedback from one or more cameras. D-SAR is challenging due to the need for real-time reconstruction of the surface shape and pose, as well as low-latency adaptive projection. To achieve this, prior works use custom hardware (e.g. high-speed, coaxial projector camera systems) [6, 10, 12] or specialized invisible markers on deformable surfaces. However, such systems do not scale to multiple projectors.

Recently, our lab has developed D-SAR systems for deformable surfaces using a Projector-Depth Camera (e.g. Kinect), or PDC unit (see Figure-1a), without requiring specialized patterns on the target surface [1–4]. We have extended this work to use multiple PDC units to achieve real-time surface shape reconstruction and

adaptive projection that conforms to the deforming surface shape. Additionally, we have developed an accurate calibration technique that registers multiple projectors and RGB-D cameras using the deformable surface itself, removing any manual calibration of the devices. These advances have the potential to enable widespread use of deformable D-SAR systems.

To demonstrate the applications of such easily deployable deformable D-SAR systems, we have explored two applications that use the system at two very different scales, showcasing its flexibility: a large multi-projector display on deformable fabrics, and a medical application for remote surgical guidance. The first application can be used in mobile and expeditionary systems where environmental factors (e.g. wind, vibrations, suction) cannot be avoided. One can create large displays on tent walls in remote, austere military or emergency operations in minutes to support large scale command and control, mission rehearsal or training operations. It can be used to create displays on mobile and inflatable objects for trade shows/events and touring entertainment applications.

For the medical application, we demonstrate our D-SAR system being used to provide remote surgical guidance for a cleft palate surgery on a cleft lip model. Projecting guidance marks right on the surgical site allows the entire surgical team to view and collaborate together. Our system can project the markings with sub-pixel accuracy, crucial for highly sensitive surgeries like the cleft lip repair in 3-6 month old infants. We will also demonstrate how the projected guidance marks remain to be bound to the surgical site even if it moves.

2 SYSTEM DESIGN

We propose to bring a D-SAR system consisting of two PDC units, a NUC mini computer to drive the system and a monitor to enable user interaction. We will demonstrate two applications: (i) a seamless display on a deformable fabric using two projectors, showcasing the system's use in large-scale displays, even in the presence of a moving surface (see Figure-1), and (ii) a surgical guidance application that will allow users to draw incision markings using a UI and project them on a model of a cleft lip in real-time (see Figure-2).

We will begin by demonstrating our automatic multi-PDC calibration method based on the work by [1]. Next, we will demonstrate our multi-projector D-SAR display on a deformable fabric, highlighting the real-time adaptive projection based on the surface shape. With the same setup, we will show the surgical assistance application by

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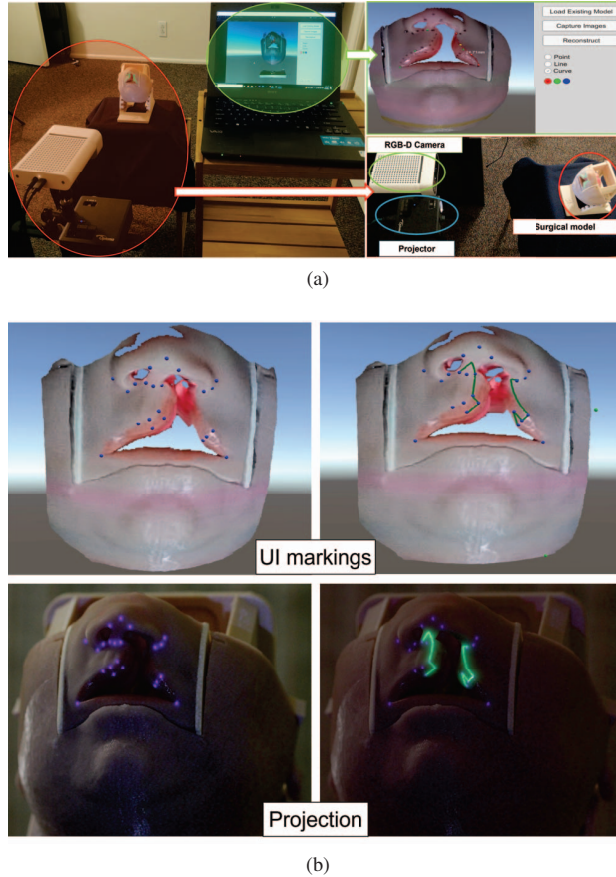


Figure 2: D-SAR for surgical guidance. (a) A PDC unit illuminating a surgical site next to a UI for drawing the markings. (b) Close-up of the markings made by a user in the UI and their corresponding projections on the surgical site.

drawing incision marks on the digital model in the UI as they get projected on the physical cleft lip model in real-time. Additionally, we will also move the physical model and demonstrate the projections quickly binding to the model accurately. We will also allow users to interact with the demos by letting them move the deformable surface, draw incisions lines on the UI and move the model. The submitted video show both demos in action and can be viewed at the following URL: <https://youtu.be/1gi12zt1biE> [5].

3 RESEARCH LAB

iGravi (Interactive Graphics and Visualization Laboratory) in University of California-Irvine, led by Prof. Aditi Majumder and Prof. M. Gopi, has spearheaded research in projection-based displays, in particular in camera-based registration of geometry and color for multi-projector displays. From 2006 till present, iGravi has advanced the frontiers of VR display design and deployment in multiple ways. We have developed inexpensive VR environments with commodity products and sophisticated and robust registration algorithms, especially for immersive non-planar display shapes including vertically extruded surfaces (e.g. cylinder), swept surfaces (e.g. truncated domes) and spherical surfaces (e.g. domes) [8, 9]. We have also developed the first complete color registration method that works better than any method available in the commercial marketplace today [7]. Next, we developed methods which allow completely uncalibrated devices in a multi-projector system on complex shapes and designed

techniques that use cross-validations and cross-references to achieve accurate and consistent geometric registration [11]. This has led to a large body of literature in multiple communities, in particular in the VGTC via multiple best paper awards. Recently, we have started exploring dynamic projection mapping on non-rigid surfaces [1–4] and are working to expand its capabilities to multiple projectors on such surfaces in the near future.

iGravi is also known for their work on computational camera and displays, computational geometry, appearance editing interactive rendering, non-photorealistic rendering and collaborative work with domains of neuroscience, medicine and ophthalmology. iGravi from UCI is a known name at many different venues including IEEE Visweek, IEEE VR, ACM Siggraph, ACM VRST, IEEE CVPR, IEEE Procams, EuroVis, Eurographics, and Interactive 3D Graphics. We have collaborators in many places including MIT, Purdue, and Disney. For the sake of brevity, we have only provided a few select references most pertinent to this demo. More information is available at <http://www.ics.uci.edu/majumder/pub.html>.

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